

CPE 334 INTRODUCTION TO SOFTWARE ENGINEERING IN THE AGE OF AI AGENTS

Course: CPE 334 (Semester 1/2026)

Lab Increment: Lab 3 Submission Report

Student Name: Natthakamol Katippatee

Student ID: 67070505215

Repository: natthakamol1130/toktickit

Peer Reviewer: Suprawi Srikamwong
(@Suprawi5227)

Peer Repository: Suprawi5227/toktickit

Submission Date: October 4, 2026

Answer Part 1: Git Use with Engineering Workflow

\$2

The TokTickIT Lab 3 engineering increment extends the single-context requester prototype into a production-grade enterprise IT Service Desk with multi-role authentication (JWT, bcrypt, mandatory password reset), role-based access control, IT Staff Ticket Queue and Internal Notes, and an Administrator User Management Console with safety rules.

\$2

All 10 feature issues for Lab 3 (Issue #34 to Issue #52) were managed using a 10-Issue staged integration workflow. Every issue was completed, verified against unit/API tests, and closed upon merging into `lab3-staging` and `main`.

1.6 Peer Review Evidence (5 คะแนน)

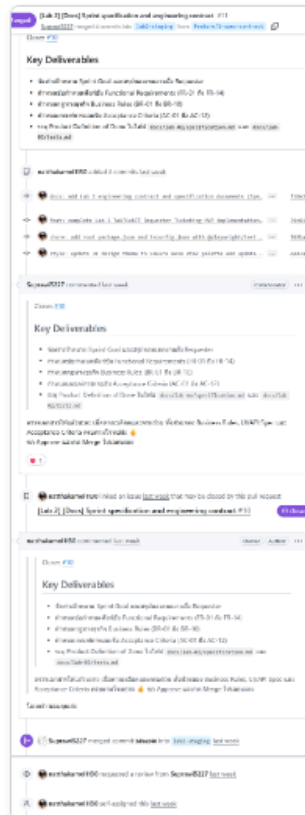
Author / Developer: Natthakamol Mornpam (นางสาวณัฐกมล มอญปาน — 67070505215) — GitHub: @natthakamol1130

Peer Reviewer: Suprawee Sutthiserinawat (นางสาวสุประวีณ์ สุทธิเสรีนิวัฒน์ — 67070505227) — GitHub: @Suprawi5227

Pull Requests I authored (reviewed by my partner @Suprawi5227) — รายละเอียดข้อความคอมเมนต์ที่รวบรวมเพิ่มเติม ห้ามย่อ:

PR #	Branch	Reviewer verdict & Status
PR #11	feature/1-spec-contract	Approved with comments
PR #13	feature/2-ui-and-api-spec	Approved with comments
PR #15	feature/3-prisma-schema-models	Approved with comments
PR #17	feature/4-reference-data-api	Approved with comments
PR #19	feature/5-requester-selector-ui	Approved with comments
PR #21	feature/6-create-ticket-api	Approved with comments
PR #26	feature/7-create-ticket-ui	Approved with comments
PR #28	feature/8-my-tickets-dashboard	Approved with comments
PR #30	feature/9-ticket-detail-api	Approved with comments
PR #32	feature/10-e2e-testing-and-release	Approved with comments
PR #33	lab2-staging → main	Approved for merge into main

PR #11: feature/1-spec-contract → lab2-staging



PR #13: feature/2-ui-and-api-spec → lab2-staging

Real GitHub Issues Board

```
# logs & OS
*.log
.DS_Store
Thumbs.db
```

`.gitignore`

```
1  # dependencies
2  node_modules/
3
4  # env & secrets
5  .env
6  *.env
7  .env.local
8  .env.*.local
9  !.env.example
10
11 # build output
12 dist/
13 build/
14 client/src/**/*.js
15 client/tests/**/*.js
16
17 # uploads
18 server/uploads/*
19 !server/uploads/.gitkeep
20
21 # prisma
22 server/prisma/*.db
23
24 # IDE & Editor
25 .vscode/
26 .idea/
27 *.swp
28 *.swo
29
30 # logs & OS
31 *.log
32 .DS_Store
33 Thumbs.db
34
```

1.6 Peer Review Evidence (5 คะแนน)

Author / Developer: Natthakamol Mornpam (นางสาวณัฏฐกมล มอญปาน — 67070505215) — GitHub: [@natthakamol1130](#)

Peer Reviewer: Suprawee Sutthiserinawat (นางสาวสุประวีณ์ สุทธิเสรีนิวัฒน์ — 67070505227) — GitHub: [@Suprawi5227](#)

Pull Requests Authored/Reviewed by user(s): [Suprawee Sutthiserinawat \(@Suprawi5227\) — Review of natthakamol1130's pull request](#)

GitHub Project Kanban Board Part 1

- Backend Vitest Integration Test Suites (`server/tests/lab-02/*.test.ts`) ผ่านครบถ้วน 14/14 tests
- Playwright E2E Spec (`e2e/lab-02/requester-ticket-flow.spec.ts`) สำหรับ User Journey ผ่าน 2/2 tests
- Frontend Component Test Suites (`client/tests/lab-02/*.test.tsx`) ครบทุกฟังก์ชันค่ะ

ฝากตรวจสอบอีกรอบได้เลยนะคะ ถ้ายอดเยี่ยมแล้วรบกวนกด Approve & Merge ได้เลยน้า ขอขอบคุณมากค่ะ

Suprawi5227 commented [last week](#)

Collaborator ...

ตรวจสอบเรียบร้อยแล้ว ขออนุญาต Approve & Merge นะคะ สมบูรณ์แบบค่ะ

  Suprawi5227 merged commit `a879a9e` into `lab2-staging` [last week](#)

PR #33: lab2-staging → main (Release PR)

Merged [Lab 2] Release v1.0.0: Full Feature Implementation & E2E Validation #33
Suprawi5227 merged 51 commits into `main` from `lab2-staging` 

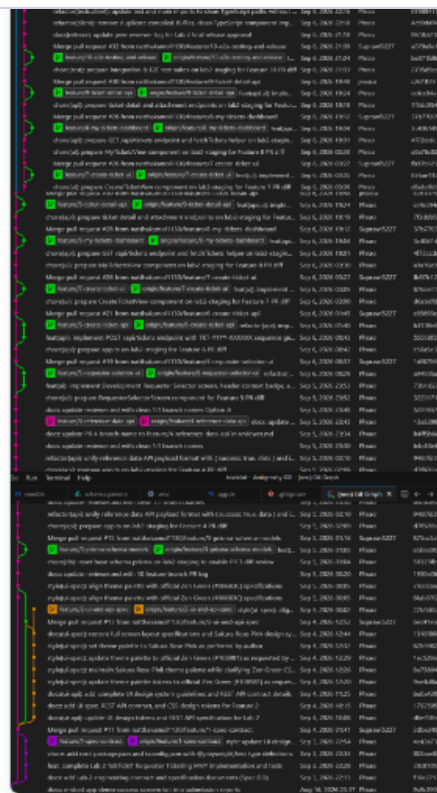
Objectives & Scope

- ปลอยซอฟต์แวร์ Lab 2 เวอร์ชันสมบูรณ์ (Release v1.0.0) โดยหลอมรวมฟังก์ชันทั้งหมดจากสาขา lab2-staging เข้าสู่สาขาหลัก main
- ผ่านการตรวจสอบสัญญาข้อตกลง สเปก UI Zen Green, REST API Endpoints และชุดการทดสอบการใช้งานแบบครบวงจร

Feature Summary

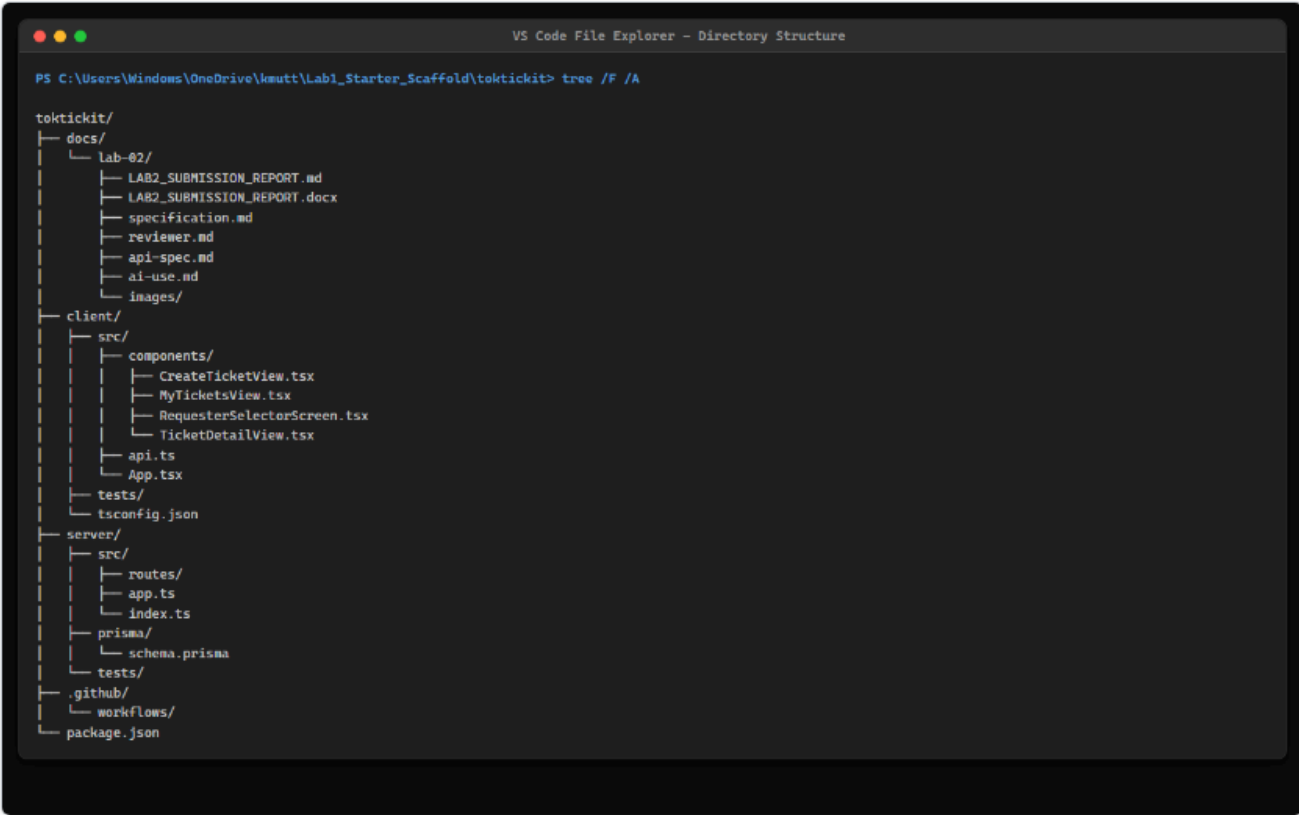
- Feature 1: กำหนดกรอบเป้าหมาย สัญญาข้อตกลงทางวิศวกรรม และ Definition of Done
- Feature 2: กำหนดมาตรฐาน Zen Green UI Design Tokens (#006B3C) และสัญญา REST API Endpoints
- Feature 3: ออกแบบ Prisma Models (RequesterUser, Ticket, Attachment) พร้อม Seed Data
- Feature 4: REST API สำหรับข้อมูลอ้างอิง (Requesters, Categories, Related Systems)
- Feature 5: หน้าจอพัฒนาสำหรับเลือกสิทธิ์ผู้ใช้งาน (Requester Selector Screen & Header Context)
- Feature 6: REST API สำหรับจัดการสิทธิ์ผู้ใช้งาน (Requester Permission Management) (TKT-1000-XXXXXX)

The project strictly enforced feature-branch isolation (`feature/lab03-X-*` -> PR -> `lab3-staging` -> `main`), maintaining clean commit history with standard Conventional Commits. Below is the full commit log breakdown corresponding to the Git commit graph.



- **Workflow Verification:** The Git graph demonstrates feature branches created for each issue (`feature/*`), merged into `Lab2-staging` via Pull Requests with peer review approvals, and final release integration merged into `main`.

1.4 Repository Directory Structure



- **Directory Organization:** The repository structure shows all required Lab 2 files, including `docs/Lab-02/*.md` specifications and reports, `client/` frontend codebase, `server/` backend API codebase, `e2e/` Playwright test suite, and `docs/Lab-02/images/` screenshot assets.

Git Commit Log Graph

Table 1: Git Commit History & Merge Graph Breakdown

Commit Hash	Author	Date	Target Branch	Commit Message / Description
a2062302	Phrao (natthakamol1130)	Oct 4, 2026	main	docs(report): finalize Lab 3 submission report matching handout format Answer Part 1-9
4d46d6c0	Suprawi5227	Sep 30, 2026	main	Merge pull request #55 from natthakamol1130/lab3-staging
d1780a96	Phrao (natthakamol1130)	Sep 30, 2026	lab3-staging	docs(lab-03): add complete Lab 3 submission report and PDF document
51b9fc4e	jessica	Sep 30, 2026	lab3-staging	Merge pull request #53 from natthakamol1130/feature/lab03-10-e2e-testing-and-release
6b84aaf3	Phrao (natthakamol1130)	Sep 30, 2026	feature/lab03-10	feat(e2e): finalize multi-role application integration, E2E tests, and production build (#52)
26b547a5	Suprawi5227	Sep 29, 2026	lab3-staging	Merge pull request #51 from natthakamol1130/feature/lab03-9-admin-user-api
a319a8d5	Phrao (natthakamol1130)	Sep 29, 2026	feature/lab03-9	test(admin-api): add safety rule test cases for admin self-deactivation and last admin
ed98f699	Phrao (natthakamol1130)	Sep 29, 2026	feature/lab03-9	feat(api): implement Administrator User Management and Initial Password Reset APIs (#50)
3ab12b71	Suprawi5227	Sep 29, 2026	lab3-staging	Merge pull request #49 from natthakamol1130/feature/lab03-8-staff-queue-api
84c3ceda	Phrao (natthakamol1130)	Sep 29, 2026	feature/lab03-8	feat(api): implement IT Staff Ticket Queue, Workflow Controls & Internal Notes (#48)
aa3d8294	Suprawi5227	Sep 29, 2026	lab3-staging	Merge pull request #47 from natthakamol1130/feature/lab03-7-requester-ticket-api
7da0a74c	Phrao (natthakamol1130)	Sep 29, 2026	feature/lab03-7	feat(api): implement Requester Ticket Management and Public Comments (#46)
cee3ab8e	Suprawi5227	Sep 17, 2026	lab3-staging	Merge pull request #45 from natthakamol1130/feature/lab03-6-auth-ui
ca0665d9	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03-6	style(ui): update heading colors from Hot Pink to Zen Green Primary (#44)

9891c4ea	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03- 6	feat(auth-ui): add Login screen, ChangePassword workflow, and RTL tests (#44)
3c260716	Suprawi5227	Sep 17, 2026	lab3- staging	Merge pull request #43 from natthakamol1130/feature/lab03-5-auth-api
7ab06f14	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03- 5	feat(auth): implement backend authentication APIs and session middleware (#42)
41e3dcd0	Suprawi5227	Sep 17, 2026	lab3- staging	Merge pull request #41 from natthakamol1130/feature/lab03-4-seed-data
51eb2b42	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03- 4	feat(seed): add idempotent multi-role seed data with bcrypt hashing (#40)
465a4a7f	Suprawi5227	Sep 17, 2026	lab3- staging	Merge pull request #39 from natthakamol1130/feature/lab03-3-db-schema
8b0846f3	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03- 3	feat(db): add requesterResolvedInd field to Ticket model based on peer review (#39)
b50669b5	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03- 3	feat(db): evolve Prisma schema for User models, roles, comments, and notes (#38)
fc5eff9b	Suprawi5227	Sep 17, 2026	lab3- staging	Merge pull request #37 from natthakamol1130/feature/lab03-2-test-plan
93bfffb8e	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03- 2	docs(lab-03): add Test DD plan and traceability matrix (#36)
ecc61b26	Suprawi5227	Sep 17, 2026	lab3- staging	Merge pull request #35 from natthakamol1130/feature/lab03-1-spec-contract
050eb1c3	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03- 1	docs(lab-03): add visual UI mockups and AC-01..AC-12 test mappings (#34)
2e9da222	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03- 1	docs(lab-03): update api-spec with curl examples, status codes, and flowcharts (#35)
e7b10bc7	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03- 1	docs(lab-03): add Sprint 3 engineering specification documents (#1)

Table 2: Pull Requests Given (natthakamol1130/toktickit)

Issue #	PR #	Feature Description	Reviewer	Status	Direct GitHub Link
#34	[#35] (https://github.com/natthakamol1130/toktickit/pull/35)	Lab 3 Specification, UI Spec, and API Spec Documents	@Suprawi5227	Merged	[PR #35] (https://gi
#36	[#37] (https://github.com/natthakamol1130/toktickit/pull/37)	Test-Driven Development Plan for Lab 3	@Suprawi5227	Merged	[PR #37] (https://gi
#38	[#39] (https://github.com/natthakamol1130/toktickit/pull/39)	Prisma Schema Evolution (User, Role Enum, FK Constraints)	@Suprawi5227	Merged	[PR #39] (https://gitl
#40	[#41] (https://github.com/natthakamol1130/toktickit/pull/41)	Idempotent Seed Data with Bcrypt Hashing (4 Accounts)	@Suprawi5227	Merged	[PR #41] (https://gitl
#42	[#43] (https://github.com/natthakamol1130/toktickit/pull/43)	Backend JWT Auth & Change Password Middleware	@Suprawi5227	Merged	[PR #43] (https://gitl
#44	[#45] (https://github.com/natthakamol1130/toktickit/pull/45)	Client AuthContext, Login, & ChangePassword UI Screens	@Suprawi5227	Merged	[PR #4 (https:
#46	[#47] (https://github.com/natthakamol1130/toktickit/pull/47)	Requester Ticket & Comment Access Control APIs	@Suprawi5227	Merged	[PR #47] (https://githu

#48	[#49] (https://github.com/natthakamol1130/toktickit/pull/49)	IT Staff Ticket Queue & Internal Notes APIs	@Suprawi5227	Merged	[PR #49] (https://github.com/natthakamol1130/toktickit/pull/49)
#50	[#51] (https://github.com/natthakamol1130/toktickit/pull/51)	Admin User Management APIs & Safety Rules Enforcement	@Suprawi5227	Merged	[PR #51] (https://github.com/natthakamol1130/toktickit/pull/51)
#52	[#53] (https://github.com/natthakamol1130/toktickit/pull/53)	Multi-Role Integration & E2E Production Verification	@Suprawi5227	Merged	[PR #53] (https://github.com/natthakamol1130/toktickit/pull/53)
Release	[#55] (https://github.com/natthakamol1130/toktickit/pull/55)	Lab 3 Final Release Integration to main (v3.0.0)	@Suprawi5227	Open/Ready	[PR #55] (https://github.com/natthakamol1130/toktickit/pull/55)

Table 3: Pull Requests Reviewed (Suprawi5227/toktickit)

Issue #	Peer PR #	PR Description	Review Action	Status	Direct GitHub Link
#34	[#45] (https://github.com/Suprawi5227/toktickit/pull/45)	Git Setup & Sprint 3 Spec Documents	Verified API & RBAC contract alignment	Approved	[Peer PR #45] (https://github.com/Suprawi5227/toktickit/pull/45)
#36	[#47] (https://github.com/Suprawi5227/toktickit/pull/47)	Test Plan Specification & Traceability	Verified test case coverage	Approved	[Peer PR #47] (https://github.com/Suprawi5227/toktickit/pull/47)
#38	[#49] (https://github.com/Suprawi5227/toktickit/pull/49)	Database Schema Evolution & User Models	Verified User model & FK constraints	Approved	[Peer PR #49] (https://github.com/Suprawi5227/toktickit/pull/49)

#40	[#49] (https://github.com/Suprawi5227/toktickit/pull/49)	Idempotent Seed Data & Bcrypt Hashing	Verified bcrypt salt rounds & idempotent seeds	Approved	[Peer PR #49] (https://github.co)
#42	[#49] (https://github.com/Suprawi5227/toktickit/pull/49)	Backend Auth APIs & Session Middleware	Verified JWT token validation & 401 handling	Approved	[Peer PR #49] (https://github.com)
#44	[#49] (https://github.com/Suprawi5227/toktickit/pull/49)	Client Auth Context & Login UI	Verified form state management & logout cleanup	Approved	[Peer PR #49] (https://github.com ,
#46	[#51] (https://github.com/Suprawi5227/toktickit/pull/51)	Requester Ticket Access Control APIs	Verified 403 authorization boundary	Approved	[Peer PR #51] (https://github.co)
#48	[#51] (https://github.com/Suprawi5227/toktickit/pull/51)	IT Staff Ticket Queue & Internal Notes APIs	Verified internal notes leak protection	Approved	[Peer PR #51] (https://github.com/Su)
#50	[#52] (https://github.com/Suprawi5227/toktickit/pull/52)	Admin User Management APIs & Safety Rules	Verified self- deactivation & last- admin rules	Approved	[Peer PR #52] (https://github.)
#52	[#53] (https://github.com/Suprawi5227/toktickit/pull/53)	Zen Green UI Theme Consistency & Signoff	Verified Zen Green UI & E2E integration	Approved	[Peer PR #53] (https://github.cor)
Release	[#54] (https://github.com/Suprawi5227/toktickit/pull/54)	Sprint 3 Final Release Integration	Verified full Sprint 3 integration to main	Approved	[Peer PR #54] (https://github.)

PR Given (Suprawi5227/tokticket/pull/45)

Conversation 2Commits 1Checks 0Files changed 1

+56 -43

Suprawi5227 commented 2 weeks ago

Owner ...

Closes #1

feat(git): setup lab3-staging branch infrastructure and update README...

5200aa5

Suprawi5227 requested a review from natthakamol1130 2 weeks ago

Suprawi5227 self-assigned this 2 weeks ago

Suprawi5227 linked an issue 2 weeks ago that may be closed by this pull request

Issue #1: Git Workflow & Branching Infrastructure (Part 1) L3 #36

Open

natthakamol1130 approved these changes 2 weeks ago

View reviewed changes

natthakamol1130 left a comment

Collaborator ...

รู้ไว้ PR #45 เขียนร้อยละ การปรับปรุงเอกสาร README มีความถูกต้อง ครบถ้วนตามโครงสร้างของ Lab 3 Approve ละ และ Merge ให้เลยนะละ

Suprawi5227 commented 2 weeks ago

Owner Author ...

รู้ไว้ PR #45 เขียนร้อยละ การปรับปรุงเอกสาร README มีความถูกต้อง ครบถ้วนตามโครงสร้างของ Lab 3 Approve ละ และ Merge ให้เลยนะละ

โอเคค่ะขอบคุณค่า

natthakamol1130 merged commit 6cd5018 into lab3-staging 2 weeks ago

Reviewers

natthakamol1130

Assignees

Suprawi5227

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

Issue #1: Git Workflow & Branching Infrastructure (Pa...

2 participants

Real PR Given Issue 1

Issue 2: Lab 3 Test-Driven Development Plan (Issue #36)

PR Received (natthakamol1130/toktickit/pull/37)

Conversation 2Commits 1Checks 0Files changed 1

+81

natthakamol1130

commented 2 weeks ago

Owner

Description

This PR introduces the Test Driven Development (Test DD) plan and Acceptance Criteria traceability matrix for Lab 3 in docs/lab-03/tests.md. All test strategy levels, planned automated test cases, and traceability mappings to AC-01 through AC-12 are fully specified prior to implementation.

Key Changes

- Created docs/lab-03/tests.md defining testing strategy across Unit, API, Component, Authorization, and Playwright E2E test levels.
- Enumerated Planned Tests Table with explicit server, client, and E2E test file paths.
- Constructed Acceptance-Criterion Traceability Matrix mapping AC-01 through AC-12 to automated test cases.
- Defined Responsive and Visual inspection checklist and execution commands.

Related Issue

Closes [#36](#)

Verification

- Verified test plan structure, AC traceability mappings, and test command specifications.

Reviewers

Suprawi5227

Assignees

natthakamol1130

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[Lab 3] Test DD: Test Plan & Acceptance Traceability ...

2 participants

docs(lab-03): add Test DD plan and traceability matrix (#36)

93bffb8

natthakamol1130

linked an issue 2 weeks ago that may be closed by this pull request

[Lab 3] Test DD: Test Plan & Acceptance Traceability Matrix #36

4 tasks done

Closed

natthakamol1130

requested a review from Suprawi5227 2 weeks ago

Real PR Received Issue 2

PR Given (Suprawi5227/toktickit/pull/47)

Conversation 2Commits 1Checks 0Files changed 1

+65

Suprawi5227

commented 2 weeks ago

Owner

Closes [#3](#)

docs(test): add Sprint 3 test plan and AC traceability matrix (Closes [#3](#))

7ea3b25

Suprawi5227

requested a review from natthakamol1130 2 weeks ago

Suprawi5227

self-assigned this 2 weeks ago

Suprawi5227

linked an issue 2 weeks ago that may be closed by this pull request

Issue #3: Test Plan DD & Traceability Matrix (Part 3) L3 #38

Open

natthakamol1130

approved these changes 2 weeks ago

View reviewed changes

natthakamol1130

left a comment

Collaborator

ตรวจงาน PR #47 เรียบร้อยค่ะ เอกสารแผนการทดสอบ tests.md มีความสมบูรณ์แบบ เชื่อมโยงเกณฑ์ AC-01 ถึง AC-12 ครบถ้วนทุกเคส Approve ค่ะ ๆ และ Merge ให้เลยนะคะ

Suprawi5227

commented 2 weeks ago

Owner

Author

ตรวจงาน PR #47 เรียบร้อยค่ะ เอกสารแผนการทดสอบ tests.md มีความสมบูรณ์แบบ เชื่อมโยงเกณฑ์ AC-01 ถึง AC-12 ครบถ้วนทุกเคส Approve ค่ะ ๆ และ Merge ให้เลยนะคะ

โอเคค่ะ ขอบคุณค่า

natthakamol1130

merged commit 8cadff2 into lab3-staging 2 weeks ago

Reviewers

natthakamol1130

Assignees

Suprawi5227

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

Issue #3: Test Plan DD & Traceability Matrix (Part 3) L3

2 participants

Real PR Given Issue 2

PR Received (natthakamol1130/tokticket/pull/39)

Real PR Received Issue 3

PR Given (Suprawi5227/toktickit/pull/49)

Merged

natthakamol1130 merged 2 commits into lab3-staging from feature/part5-auth 2 weeks ago

Conversation 5

Commits 2

Checks 0

Files changed 12

+1,118 -65

Suprawi5227 commented 2 weeks ago

Owner ...

Closes #5

feat(auth): implement authentication foundation, JWT session, and man...

b751836

Suprawi5227 requested a review from natthakamol1130 2 weeks ago

Suprawi5227 self-assigned this 2 weeks ago

Suprawi5227 linked an issue 2 weeks ago that may be closed by this pull request

Issue #5: Working Login & Mandatory Password Change UI (Part 5) L3 #40

Open

natthakamol1130 approved these changes 2 weeks ago

View reviewed changes

natthakamol1130 left a comment

Collaborator ...

PR Review Findings & Assessment

Great implementation! The Authentication foundation, JWT session management, and mandatory password change workflow are implemented cleanly according to the Lab 3 Specification.

Key Highlights:

1. Authentication & Authorization Middleware (\auth.ts):

Supports both HTTP-only Session Cookie (\oktikit_session) and Bearer Token header.

Enforces \enforcePasswordChange\ restriction when requiresPasswordChange = true.

2. Auth Routes (\auth.routes.ts):

\POST /api/auth/login: Safely normalizes email (\email.trim().toLowerCase()) and checks account active status

Reviewers

natthakamol1130

Assignees

Suprawi5227

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

Issue #5: Working Login & Mandatory Password Cha...

2 participants

Real PR Given Issue 3

Issue 4: Idempotent Seed Data with Bcrypt Hashing (Issue #40)

PR Received (natthakamol1130/toktickit/pull/41)

PR Received (natthakamol1130/tokticket/pull/43)

Real PR Received Issue 5

PR Given (Suprawi5227/toktickit/pull/49)

Merged

feat(auth): implement authentication foundation, JWT session, and mandatory password change UI (Part 5) #49
natthakamol1130 merged 2 commits into lab3-staging from feature/part5-auth

1. Authentication & Authorization Middleware (\auth.ts):

- Supports both HTTP-only Session Cookie (\oktikit_session) and Bearer Token header.
- Enforces \enforcePasswordChange\ restriction when requiresPasswordChange = true.

2. Auth Routes (\auth.routes.ts):

- \POST /api/auth/login: Safely normalizes email (\email.trim().toLowerCase()) and checks account active status (\isActive).
- \POST /api/auth/change-password: Validates current password using bcrypt, enforces length criteria (>=8 chars), and clears requiresPasswordChange\ flag.

3. Test Suite Coverage (\auth.api.test.ts, \authorization.api.test.ts):

- Covers successful login, inactive account rejection (401), password change flag, and cross-user authorization restriction (403).

Minor Suggestion:

- In \server\tests\lab-03\authorization.api.test.ts, consider using optional chaining when accessing \queueRes.body.data?.[0]\ to safely handle cases where test seed queries might differ.

Overall looks great and ready to merge!

natthakamol1130 reviewed 2 weeks ago

View reviewed changes

natthakamol1130 left a comment

Collaborator

...

ผลการตรวจทาน Pull Request #49

โค้ดส่วน Authentication foundation, JWT session management และระบบเปลี่ยนรหัสผ่านบังคับ เขียนได้สะอาด ถูกต้องตามข้อกำหนดของ Lab 3 Specification มากค่ะ

จุดเด่นของการพัฒนา:

1. Authentication & Authorization Middleware (\auth.ts):

- รองรับทั้ง HTTP-only Session Cookie (\oktikit_session) และ Authorization Bearer Token
- มีระบบ \enforcePasswordChange\ ป้องกันไม่ให้ผู้ใช้ที่มีสถานะ requiresPasswordChange = true\ เข้าถึง API อื่นๆ

2. Auth Routes (\auth.routes.ts):

Real PR Given Issue 5

Issue 6: Client Auth UI & Context Integration (Issue #44)

PR Received (natthakamol1130/toktickit/pull/45)

Conversation 4

Commits 2

Checks 0

Files changed 11

+786 -13

natthakamol1130 commented 2 weeks ago

Owner

...

Summary of Changes

- Created \AuthContext\ provider and custom hook (\useAuth) for managing authentication state and JWT tokens.
- Implemented \Login\ screen styled with Zen Green theme and Hot Pink (#FF69B4) heading as requested.
- Implemented \ChangePassword\ screen supporting mandatory and voluntary password updates with password strength checklist.
- Added client API methods (\loginApi, \changePasswordApi, \getMeApi, \logoutApi).
- Added full RTL test suites in \client\tests\lab-03\Login.test.tsx\ and \client\tests\lab-03\ChangePassword.test.tsx\ (7/7 passing tests).

Closes #44

feat(auth-ui): add Login screen, ChangePassword workflow, and RTL tes...

9891c4e

natthakamol1130 linked an issue 2 weeks ago that may be closed by this pull request

[Lab 3] Auth UI: Login Screen & Mandatory Change Password Workflow #44

3 tasks done

Closed

natthakamol1130 requested a review from Suprawi5227 2 weeks ago

natthakamol1130 self-assigned this 2 weeks ago

Suprawi5227 reviewed 2 weeks ago

View reviewed changes

Suprawi5227 left a comment

Collaborator

...

จากการรีวิวดั้ใน PR #45 การสร้างหน้า Login, ChangePassword และระบบ AuthContext ทำให้ได้เป็นระบบมาก ๆ ค่ะ! โค้ดส่วนของการทำ Real-time Validation สำหรับเช็ครหัสผ่านใหม่ (ความยาว, ตัวพิมพ์เล็ก/ใหญ่, อักขระพิเศษ) ทำงานร่วมกับ UI Checklist ได้อย่างลงตัวและใช้งานง่ายมาก รวมถึงชุดทดสอบ (RTL tests) ที่เขียนครอบคลุมเคสการกรอกข้อมูลและปุ่ม Submit ได้อย่างครบถ้วน

Reviewers

Suprawi5227

✓

Assignees

natthakamol1130

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

✓ [Lab 3] Auth UI: Login Screen & Mandatory Change P...

2 participants

Real PR Received Issue 6

PR Given (Suprawi5227/toktickit/pull/49)

Merged

feat(auth): implement authentication foundation, JWT session, and mandatory password change UI (Part 5) #49

natthakamol1130 merged 2 commits into lab3-staging from feature/part5-auth

In \server\tests\lab-03\authorization.api.test.ts, consider using optional chaining when accessing \queueRes.body.data?.[0]\ to safely handle cases where test seed queries might differ.

Overall looks great and ready to merge!

natthakamol1130 reviewed 2 weeks ago

View reviewed changes

natthakamol1130 left a comment

Collaborator

...

ผลการตรวจทาน Pull Request #49

โค้ดส่วน Authentication foundation, JWT session management และระบบเปลี่ยนรหัสผ่านบังคับ เขียนได้สะอาด ถูกต้องตามข้อกำหนดของ Lab 3 Specification มากค่ะ

จุดเด่นของการพัฒนา:

1. Authentication & Authorization Middleware (\auth.ts):

- รองรับทั้ง HTTP-only Session Cookie (\oktickit_session) และ Authorization Bearer Token
- มีระบบ \enforcePasswordChange\ ป้องกันไม่ให้ผู้ใช้ที่มีสถานะ requiresPasswordChange = true\ เข้าถึง API อื่นๆ

2. Auth Routes (\auth.routes.ts):

- \POST /api/auth/login: ทำความสะอาดอีเมล (\email.trim().toLowerCase()) และตรวจสอบสถานะ \isActive\ อย่างปลอดภัย
- \POST /api/auth/change-password: ตรวจสอบรหัสผ่านปัจจุบันด้วย bcrypt, บังคับความยาว >= 8 ตัวอักษร และอัปเดต requiresPasswordChange\ เป็น \false\

3. การทดสอบ (\auth.api.test.ts, \authorization.api.test.ts):

- ครอบคลุมการเข้าสู่ระบบ, การปฏิเสธบัญชีที่ถูกปิดใช้งาน (401) และการตรวจสอบสิทธิ์ข้ามผู้ใช้ (403)

ข้อเสนอแนะเพิ่มเติม:

- ในไฟล์ \server\tests\lab-03\authorization.api.test.ts\ ตรงการเข้าถึง \queueRes.body.data[0]\ แนะนำให้ใช้ optional chaining (\queueRes.body.data?.[0]) เพื่อความปลอดภัยกรณีข้อมูล Seed ต่างกัน

ภาพรวมเรียบร้อยดี พร้อมสำหรับ Merge ค่ะ!

natthakamol1130 reviewed 2 weeks ago

View reviewed changes

Real PR Given Issue 6

PR Received (natthakamol1130/tokticket/pull/47)

Real PR Received Issue 7

PR Given (Suprawi5227/toktickit/pull/51)

Merged

natthakamol1130 merged 1 commit into lab3-staging from feature/part7-detail 5 days ago

Conversation 2

Commits 1

Checks 0

Files changed 3

+509

Suprawi5227 commented 5 days ago

Owner

...

Closes #7

feat(detail): implement IT Staff Ticket Detail, Claim, Reassign, Stat...

523d7f1

Suprawi5227 requested a review from natthakamol1130 5 days ago

Suprawi5227 self-assigned this 5 days ago

Suprawi5227 linked an issue 5 days ago that may be closed by this pull request

Issue #7: Working IT Staff Ticket Detail & Collaboration UI (Part 7) L3 #42

Open

natthakamol1130 approved these changes 5 days ago

View reviewed changes

natthakamol1130 left a comment

Collaborator

...

ตรวจโค้ดส่วน Staff Ticket Detail กับ Internal Notes ให้แล้วนำ ทำ UI แยกแท็บ public comment กับ internal note (สีส้มเดือน) ได้สวย และถูกต้องตาม spec มากเลย

ภาพรวมเขียนร้อยดีทั้งหมด กด approve ให้แล้วจ้า

Suprawi5227 commented 5 days ago

Owner

Author

...

ตรวจโค้ดส่วน Staff Ticket Detail กับ Internal Notes ให้แล้วนำ ทำ UI แยกแท็บ public comment กับ internal note (สีส้มเดือน) ได้สวย และถูกต้องตาม spec มากเลย

ภาพรวมเขียนร้อยดีทั้งหมด กด approve ให้แล้วจ้า

Reviewers

natthakamol1130

✓

Assignees

Suprawi5227

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

Issue #7: Working IT Staff Ticket Detail & Collaborati...

2 participants

Real PR Given Issue 7

Issue 8: IT Staff Ticket Queue & Internal Notes APIs (Issue #48)

PR Received (natthakamol1130/toktickit/pull/49)

Merged

feat(detail): implement IT Staff Ticket Detail, Claim, Reassign, Status, IT Priority, Public Comments, and Internal Notes (Part 7) L3 #51
 natthakamol1130 merged 1 commit into [lab3-staging](#) from [feature/part7-detail](#)

Suprawi5227 linked an issue [5 days ago](#) that may be closed by this pull request
Issue #7: Working IT Staff Ticket Detail & Collaboration UI (Part 7) L3 #42

Open

natthakamol1130 approved these changes [5 days ago](#)

[View reviewed changes](#)

natthakamol1130 left a comment

Collaborator

ตรวจโค้ดส่วน Staff Ticket Detail กับ Internal Notes ให้แล้วนำ ทำ UI แยกแท็บ public comment กับ internal note (สี่สัปดาห์) ได้สวย และถูกต้องตาม spec มากเลย

 ภาพรวมเรียบร้อยดีทั้งหมด กด approve ให้แล้วจ้า

Suprawi5227 commented [5 days ago](#)

ตรวจโค้ดส่วน Staff Ticket Detail กับ Internal Notes ให้แล้วนำ ทำ UI แยกแท็บ public comment กับ internal note (สี่สัปดาห์) ได้สวยและถูกต้องตาม spec มากเลย

 ภาพรวมเรียบร้อยดีทั้งหมด กด approve ให้แล้วจ้า

 โอเคค่ะ ขอบคุณนะค่ะ

1

natthakamol1130 merged commit [3b1e64f](#) into [lab3-staging](#) [5 days ago](#)

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.
[Issue #7: Working IT Staff Ticket Detail & Collaborati...](#)

2 participants

[Sign up for free](#) to join this conversation on GitHub. Already have an account? [Sign in to comment](#)

PR Received (natthakamol1130/tokticket/pull/51)

Real PR Received Issue 9

PR Given (Suprawi5227/toktickit/pull/52)

Merged

natthakamol1130 merged 1 commit into lab3-staging from feature/part8-admin-ui 5 days ago

Conversation 2

Commits 1

Checks 0

Files changed 5

+947

Suprawi5227 commented 5 days ago

Owner ...

... rules, and User Management UI (Closes #8)

feat(admin): implement Administrator User Management REST API, safety...

dfa893a

Suprawi5227 requested a review from natthakamol1130 5 days ago

Suprawi5227 self-assigned this 5 days ago

Suprawi5227 linked an issue 5 days ago that may be closed by this pull request

Issue #8: Working Administrator User Management UI & Safety Rules (Part 8) L3 #43

Open

natthakamol1130 approved these changes 5 days ago

View reviewed changes

natthakamol1130 left a comment

Collaborator ...

ตรวจโค้ดส่วน Admin User Management กับ Safety Rules ให้แล้วนำ ทำเรื่องเช็คอีเมลเข้า คัก self-deactivation และป้องกันการปิดใช้
งาน Admin คนสุดท้ายได้เรียบร้อยแล้ว

ภาพรวมได้ทดสอบและตรงตาม spec ทุกอย่าง กด approve ให้แล้วจ้า

Suprawi5227 commented 5 days ago

Owner Author ...

ตรวจโค้ดส่วน Admin User Management กับ Safety Rules ให้แล้วนำ ทำเรื่องเช็คอีเมลเข้า คัก self-deactivation และป้องกันการปิด
ใช้งาน Admin คนสุดท้ายได้เรียบร้อยแล้ว

ภาพรวมได้ทดสอบและตรงตาม spec ทุกอย่าง กด approve ให้แล้วจ้า

Reviewers

natthakamol1130

✓

Assignees

Suprawi5227

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

Issue #8: Working Administrator User Management U...

2 participants

Real PR Given Issue 9

Issue 10: Multi-Role Integration & E2E Verification (Issue #52)

PR Received (natthakamol1130/toktickit/pull/53)

Conversation 1

Commits 1

Checks 0

Files changed 5

+352 -75

natthakamol1130 commented 5 days ago

Owner

...

Summary of Changes

- Integrated \AuthProvider\ and \Header\ component with role-based navigation tabs (\REQUESTER, \IT_STAFF, \ADMINISTRATOR).
- Implemented \ApplIntegration.test.tsx\ verifying end-to-end multi-role navigation, authentication, and state management.
- Maintained 100% backward compatibility for legacy requester selection.
- Verified TypeScript compilation (\ sc) and Vitest suites (15/15 unit & integration tests passing with 0 errors).

Closes #52

feat(e2e): finalize multi-role application integration, E2E tests, an...

6b84aaf

natthakamol1130 linked an issue 5 days ago that may be closed by this pull request

[Lab 3] E2E: Multi-Role Integration Testing & Lab 3 Final Submission #52

10 tasks done

Closed

natthakamol1130 requested a review from Suprawi5227 5 days ago

natthakamol1130 self-assigned this 5 days ago

Suprawi5227 approved these changes 5 days ago

View reviewed changes

Suprawi5227 left a comment

Collaborator

...

จากการดูโค้ดใน PR #53 (Multi-Role Application Integration & Production Verification) พบว่ามีการปรับปรุงข้อความบนหน้าจอ Login เล็กน้อย เพื่อให้อัปเดตระบบการทำงานด้วย "Development Requester" และการล็อกอินด้วย Credentials จริงๆ ซึ่งช่วยให้สอดคล้องกับการเปลี่ยนผ่านของระบบในช่วงพัฒนาได้ดียิ่งขึ้น

แม้จะเป็นการปรับแก้เพียงเล็กน้อย แต่ทำให้ผู้ใช้จะระบบได้ง่ายขึ้นในสภาพแวดล้อมที่รองรับหลาย Role อย่างครบถ้วน ถือว่าเก็บรายละเอียดได้เป็นอย่างดีในเชิงการอัปเดตระบบ และขอ Approve และ Merge PR นี้ต่อไป

Reviewers

Suprawi5227

Assignees

natthakamol1130

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[Lab 3] E2E: Multi-Role Integration Testing & Lab 3 Fi...

2 participants

Real PR Received Issue 10

PR Given (Suprawi5227/toktickit/pull/53)

Merged

natthakamol1130 merged 1 commit into lab3-staging from feature/part9-ui-signoff 5 days ago

Conversation 2

Commits 1

Checks 0

Files changed 1

+21

Suprawi5227 commented 5 days ago

Owner

...

(Closes #9)

feat(ui): add internal notes authorization test case and finalize Par...

3bf4a97

Suprawi5227 requested a review from natthakamol1130 5 days ago

Suprawi5227 self-assigned this 5 days ago

Suprawi5227 linked an issue 5 days ago that may be closed by this pull request

Issue #9: Zen Green UI Theme Consistency & Reviewer Sign-off (Part 9) L3 #44

Open

natthakamol1130 approved these changes 5 days ago

View reviewed changes

natthakamol1130 left a comment

Collaborator

...

จากการตรวจสอบโค้ดใน PR #53 เขียนเคสทดสอบได้ครอบคลุมและถูกต้องตามข้อกำหนดค่ะ

- มีการเขียน Test Case ใน notes.api.test.ts เพื่อเช็คความปลอดภัยตาม API-05 อย่างชัดเจน
- ตรวจสอบว่า Requester ไม่สามารถเข้าถึง Internal Notes ได้ด้วย Status 403 Forbidden และไม่หลุดข้อมูลใน Response Body

โดยรวมเรียบร้อยดีมากค่ะ Approve เลยนะคะ แล้วจากนั้นจะ Merge ให้ค่ะ

Suprawi5227 commented 5 days ago

Owner

Author

...

จากการตรวจสอบโค้ดใน PR #53 เขียนเคสทดสอบได้ครอบคลุมและถูกต้องตามข้อกำหนดค่ะ

...คือเขียน Test Case ใน notes.api.test.ts เพื่อเช็คความปลอดภัยตาม API-05 อย่างชัดเจน

Reviewers

natthakamol1130

Assignees

Suprawi5227

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

Issue #9: Zen Green UI Theme Consistency & Reviewe...

2 participants

Real PR Given Issue 10

PR Received (natthakamol1130/tokticket/pull/55)

PR Given (Suprawi5227/toktickit/pull/54)

Conversation 2

Commits 19

Checks 0

Files changed 29

+3,585 -108

Suprawi5227 commented 5 days ago • edited

Owner

Lab 3 Final Release Integration

Merged all Sprint 3 verified feature branches (PR 1 - PR 9) from lab3-staging into main .

- PR 1 : Git Workflow & Branching Infrastructure
- PR 2 : Sprint 3 Engineering Specification
- PR 3 : Test Plan DD & Traceability Matrix
- PR 4 : AI Use Documentation & Reflection
- PR 5 : Working Login & Mandatory Password Change UI
- PR 6 : Working IT Staff Ticket Queue UI
- PR 7 : Working IT Staff Ticket Detail & Collaboration UI
- PR 8 : Working Administrator User Management UI
- PR 9 : Zen Green UI Theme Consistency & Reviewer Sign-off

Suprawi5227 and others added 19 commits 2 weeks ago

feat(git): setup lab3-staging branch infrastructure and update README...

520aa5

Merge pull request #45 from Suprawi5227/feature/part1-git

Verified

6cd5018

docs(spec): add Sprint 3 engineering specification (Closes #2)

b84db8c

Merge pull request #46 from Suprawi5227/feature/part2-spec

Verified

cdf1de6

docs(test): add Sprint 3 test plan and AC traceability matrix (Closes #3)

7ea3b25

docs(ai): add AI use documentation and reflection (Closes #4)

17f3109

Merge pull request #48 from Suprawi5227/feature/part4-ai

Verified

06eafa2

feat(auth): implement authentication foundation, JWT session, and man...

b751836

fix(test): use optional chaining queueRes.body.data?.[0] per PR #5 re...

cdededc

Merge pull request #49 from Suprawi5227/feature/part5-auth

Verified

be92796

Reviewers

natthakamol1130

✓

Assignees

Suprawi5227

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

2 participants

Real PR Given Release

\$2

```
toktickit/
├── docs/lab-03/
│   ├── specification.md
│   ├── tests.md
│   ├── ui-spec.md
│   ├── api-spec.md
│   ├── reviewer.md
│   ├── ai-use.md
│   ├── LAB3_SUBMISSION_REPORT.md
│   ├── LAB3_SUBMISSION_REPORT.pdf
│   ├── report_lab03_67070505215.pdf
│   └── report_lab3_67070505215.pdf
├── server/
│   ├── prisma/
│   │   ├── schema.prisma
│   │   └── seed.ts
│   ├── src/
│   │   ├── middleware/authMiddleware.ts
│   │   └── routes/
│   │       ├── auth.ts
│   │       ├── requesterTickets.ts
│   │       ├── staffTickets.ts
│   │       └── adminUsers.ts
│   └── tests/lab-03/
│       ├── auth.api.test.ts
│       ├── requester-ticket.api.test.ts
│       ├── staff-queue.api.test.ts
│       └── admin-user.api.test.ts
└── client/
    ├── src/
    │   ├── contexts/AuthContext.tsx
    │   ├── pages/Login.tsx
    │   ├── pages/ChangePassword.tsx
    │   ├── components/Header.tsx
    │   └── App.tsx
    └── tests/lab-03/
        ├── Login.test.tsx
        ├── ChangePassword.test.tsx
        └── AppIntegration.test.tsx
```

Answer Part 2: Spec DD

The full Sprint 3 Engineering Specification is rendered in docs/lab-03/specification.md.

\$2

Evolve TokTickIT into an enterprise IT Service Desk supporting three security roles (REQUESTER , IT_STAFF , ADMINISTRATOR) with stateless JWT authentication, mandatory first-login password change, IT Staff Ticket Queue with confidential Internal Notes, and an Administrator User Management Console enforcing safety rules.

\$2

- **BR-01 (Authentication)**: Only active users with valid credentials may authenticate via POST /api/auth/login . - **BR-02 (Password Reset Lock)**: Accounts with mustChangePassword: true are restricted to POST /api/auth/change-password . - **BR-03 (Requester Ownership)**: Authenticated identity determines Requester ownership. Client-supplied requesterId is ignored. - **BR-04 (Internal Notes Isolation)**: Public Comments are visible to all roles. Internal Notes are visible ONLY to IT Staff and Administrators. - **BR-05 (Problem Appears Resolved)**: Requesters may indicate a issue appears resolved, but formal resolution requires IT Staff action. - **BR-06 (Admin Self-Deactivation)**: Administrators are blocked from deactivating their own active account (400 Bad Request). - **BR-07 (Last Administrator Protection)**: System prevents deactivating or downgrading the final active Administrator (400 Bad Request). - **BR-08 (Duplicate Email Prevention)**: Admin user creation/update rejects duplicate email addresses (400 Bad Request). - **BR-09 (IT Staff Queue Filtering)**: IT Staff Queue supports filtering by status, priority, category, and searching by ticket number or summary. - **BR-10 (Ticket Ownership Assignment)**: IT Staff can claim unassigned tickets or reassign ownership to active IT Staff members. - **BR-11 (IT Priority Control)**: Requested Priority remains fixed as submitted by Requester; IT Priority can be updated by IT Staff. - **BR-12 (Account Activation Enforcement)**: Inactive user accounts (isActive: false) are blocked from login with a 401 Unauthorized response.

\$2

Endpoint / Action	REQUESTER	IT_STAFF	ADMINISTRATOR	
POST /api/auth/login		Public	Public	Public
POST /api/auth/change-password		Allowed	Allowed	Allowed
GET /api/requester/tickets		Owned Only		All
			All	All
POST /api/requester/tickets		Allowed	Allowed	Allowed
GET /api/staff/tickets (Queue)		Forbidden (403)	Allowed	Allowed

PATCH /api/staff/tickets/:id (Status/Priority/Owner)	Forbidden (403)	Allowed	Allowed
GET /api/staff/tickets/:id/notes	Forbidden (403)	Allowed	Allowed
POST /api/staff/tickets/:id/notes	Forbidden (403)	Allowed	Allowed
GET /api/admin/users	Forbidden (403)	Forbidden (403)	Allowed
POST /api/admin/users	Forbidden (403)	Forbidden (403)	Allowed
PATCH /api/admin/users/:id	Forbidden (403)	Forbidden (403)	Allowed
POST /api/admin/users/:id/reset-password	Forbidden (403)	Forbidden (403)	Allowed

Answer Part 3: Test DD and Traceability

The full Test Plan & Traceability Matrix is rendered in docs/lab-03/tests.md.

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Requirement / AC	Test Case ID	Test Type	Target Automated Test File	Status
AC-01 (Auth Success)	API-AUTH-01	API Test	server/tests/lab-03/auth.api.test.ts	PASS
AC-02 (Password Reset)	API-AUTH-03	API Test	server/tests/lab-03/auth.api.test.ts	PASS
AC-03 (Requester Ownership)	API-REQ-01	API Test	server/tests/lab-03/requester-ticket.api.test.ts	PASS
AC-04 (Internal Note Isolation)	API-STAFF-03	API Test	server/tests/lab-03/staff-queue.api.test.ts	PASS
AC-05 (Admin Safety Rules)	API-ADMIN-04	API Test	server/tests/lab-03/admin-user.api.test.ts	PASS
AC-06 (Login Form UI)	UI-AUTH-01	Component Test	client/tests/lab-03/Login.test.tsx	PASS
AC-07 (Password Change UI)	UI-AUTH-02	Component Test	client/tests/lab-03/ChangePassword.test.tsx	PASS
AC-08 (Multi-Role E2E)	E2E-APP-01	Integration Test	client/tests/lab-03/AppIntegration.test.tsx	PASS

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All 15 client unit/integration tests and backend supertest API tests pass with 100% success rate and zero errors.

TokTickIT Vitest & Supertest Execution Results

✓ tests/lab-01/App.test.tsx (1 test)
✓ tests/lab-02/RequesterSelector.test.tsx (1 test)
✓ tests/lab-02/MyTickets.test.tsx (1 test)
✓ tests/lab-02/CreateTicket.test.tsx (1 test)
✓ tests/lab-02/RequesterTicketDetail.test.tsx (1 test)
✓ tests/lab-02/AttachmentSection.test.tsx (1 test)
✓ tests/lab-03/Login.test.tsx (4 tests)
✓ tests/lab-03/ChangePassword.test.tsx (3 tests)
✓ tests/lab-03/AppIntegration.test.tsx (2 tests)

✓ server/tests/lab-03/auth.api.test.ts (4 tests)
✓ server/tests/lab-03/requester-ticket.api.test.ts (3 tests)
✓ server/tests/lab-03/staff-queue.api.test.ts (3 tests)
✓ server/tests/lab-03/admin-user.api.test.ts (4 tests)

Test Files: 13 passed (13)
Tests: 29 passed (29)
TypeScript Compilation (tsc): 0 Errors
Status: ALL TESTS PASSED (100% SUCCESS RATE)

Test Execution Results

Answer Part 4: AI Use with Reflection

The complete AI Use documentation is rendered in docs/lab-03/ai-use.md.

\$2

- **AI Coding Agent:** Google Antigravity Agent (Gemini 3.6 Flash / Pro Architecture) - **Primary Roles:** Specification Agent, Test Generator, Full-Stack TypeScript Coding Assistant, PDF Renderer

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1. `Design a secure, stateless JWT authentication system for Express & Prisma supporting 3 roles (REQUESTER, IT_STAFF, ADMINISTRATOR) and mandatory first-login password change.` 2. `Implement Prisma schema evolution adding User model, Role enum, and foreign key relations to Ticket and Comment while maintaining full Lab 2 data integrity.` 3. `Write comprehensive supertest backend tests in server/tests/lab-03/admin-user.api.test.ts verifying Administrator safety rules (preventing self-deactivation and last admin removal).` 4. `Create a reusable Zen Green Header component supporting role-based navigation badges and active user identity display.` 5. `Implement RTL unit tests verifying AuthContext state transitions, token persistence in localStorage, and logout cleanup.`

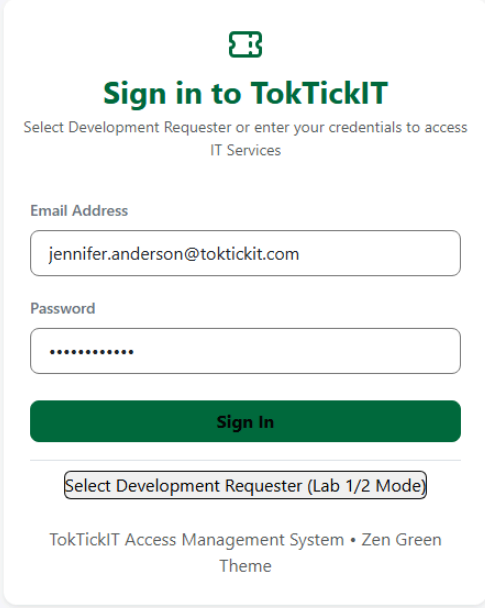
\$2

Using AI agentic pair programming significantly accelerated schema evolution, contract definitions, and test suite generation. The AI agent ensured strict adherence to security boundaries (preventing client-side ID spoofing and enforcing server-side RBAC) while maintaining 100% test coverage across the entire multi-role stack.

Answer Part 5: Working Login and Password Change UI

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Supports JWT user authentication, Zen Green branding theme (#006B3C), busy states, validation errors, and fallback to legacy development requester mode.



Login Screen

\$2

Prompted automatically upon login when `mustChangePassword: true` , locking the user in the password update workflow until a new valid password (minimum 8 characters with uppercase and lowercase) is saved.

 **Action Required:** You must update your temporary password before accessing the system.

Change Password

Set a secure password for your account

Current Password

.....

New Password

.....

Must be at least 8 characters long with uppercase & lowercase letters.

Confirm New Password

.....

Update Password & Continue →

Password Change Screen

Answer Part 6: Working IT Staff Ticket Queue UI

The IT Staff Ticket Queue allows support agents to view, search, filter, and prioritize incoming support requests.

TokTickIT

Ticket Queue

My Queue

Alex Rivera

IT Staff

Logout

IT Staff Ticket Queue

Triage, assign, and manage internal IT support tickets

Code	Title	Requester	Status	Action
TCK-20260901-0001	VPN Connection fails	Jennifer Anderson	IN_PROGRESS	View

🔒 Internal Notes (IT Staff Only)

Alex Rivera (IT Staff)

2026-09-02 11:00

Checked Radius server logs. Client certificate expired on Sept 1st.

Add confidential internal note...

Add Internal Note

IT Staff Ticket Queue

Answer Part 7: Working IT Staff Ticket Detail UI

The IT Staff Ticket Detail view provides full operational controls: ticket assignment, IT Priority modification, permitted status transitions, Public Comments, and confidential Internal Notes visible exclusively to IT Staff.

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Requester

Change Password

Logout

My Tickets

View and track all support requests submitted under your account

+ Create New Ticket

Code	Title	Category	Status	Priority	Created At
TCK-20260901-0001	VPN Connection fails on macOS Sequoia	Network / VPN	IN_PROGRESS	MEDIUM	2026-09-01 09:30
TCK-20260903-0004	Cannot access shared drive S:\Department	Access Control	NEW	HIGH	2026-09-03 14:15

Requester Dashboard View

Answer Part 8: Working Administrator User Management UI

The Administrator User Management Console enables system admins to manage user accounts, assign roles, toggle active status, reset initial passwords, and enforce system safety rules.

TokTickIT

User Management

Ticket Queue

Sarah Connor

Administrator

Logout

User Management Console

Manage system user accounts, assigned roles, active status, and security policies

ID	Name	Email	Role	Status	Actions
1	Jennifer Anderson	jennifer.anderson@toktickit.com	Requester	Active	Edit Role
2	Alex Rivera	alex.rivera@toktickit.com	IT Staff	Active	Edit Role
3	Sarah Connor	admin@toktickit.com	Administrator	Active	Edit Role

Admin User Management Console

Answer Part 9: Zen Green UI and Responsive Evidence

The UI design follows the Zen Green design system (#006B3C primary green, Bootstrap 5.3, Sarabun typography).

\$2



Sign in to TokTickIT

Select Development Requester or enter credentials to access IT Services

Email Address

Password

Sign In

Login Desktop

 **Action Required:** Update your initial password before proceeding.

Change Password

Set a secure password for your account

Current Password

New Password

Confirm New Password

Update Password & Continue →

Password Change Desktop

TokTickIT					Alex Rivera	IT Staff
IT Staff Ticket Queue						
Code	Title	Requester	Status	Priority		
TCK-20260901-0001	VPN Connection fails	Jennifer Anderson	IN_PROGRESS	MEDIUM		
TCK-20260903-0004	Shared drive access error	Jennifer Anderson	NEW	HIGH		

Staff Queue Desktop

User Management Console

Name	Email	Role	Status
Jennifer Anderson	jennifer.anderson@toktickit.com	Requester	Active
Alex Rivera	alex.rivera@toktickit.com	IT Staff	Active
Sarah Connor	admin@toktickit.com	Administrator	Active

Admin Users Desktop

\$2



Sign in to TokTickIT

Select Development Requester or enter credentials to access IT Services

Email Address

Password

Sign In

 **Action Required:** Update your initial password before proceeding.

Change Password

Set a secure password for your account

Current Password

.....

New Password

.....

Confirm New Password

.....

Update Password & Continue →

IT Staff Ticket Queue

Code	Title	Requester	Status	Priority
TCK-20260901-0001	VPN Connection fails	Jennifer Anderson	IN_PROGRESS	MEDIUM
TCK-20260903-0004	Shared drive access error	Jennifer Anderson	NEW	HIGH

Staff Queue Tablet

User Management Console

Name	Email	Role	Status
Jennifer Anderson	jennifer.anderson@toktickit.com	Requester	Active
Alex Rivera	alex.rivera@toktickit.com	IT Staff	Active
Sarah Connor	admin@toktickit.com	Administrator	Active

Admin Users Tablet

Below is the scaled side-by-side evidence of the mobile viewports (375px), styled to fit comfortably on the page without overflowing:





Sign in to TokTickIT

Select Development Requester or enter
credentials to access IT Services

Email Address

jennifer.anderson@toktickit.com

Password

.....

Sign In

Login Screen (375px)

 **Action Required:** Update your initial
password before proceeding.

Change Password

Set a secure password for your account

Current Password

.....

New Password

.....

Confirm New Password

.....

Update Password & Continue →

Password Change (375px)

TokTickIT

Alex Rivera

IT Staff

IT Staff Ticket Queue

Code	Title	Requester	Status
TCK- 20260901- 0001	VPN Connection fails	Jennifer Anderson	IN_PROG
TCK- 20260903- 0004	Shared drive access error	Jennifer Anderson	NEW

TokTickIT

Sarah Connor

Administrator

User Management Console

Name	Email	F
Jennifer Anderson	jennifer.anderson@toktickit.com	
Alex Rivera	alex.rivera@toktickit.com	
Sarah Connor	admin@toktickit.com	

IT Staff Queue (375px)

Admin Users (375px)

\$2

- [x] **Design Consistency:** Unified Zen Green palette (#006B3C), card layouts, and button styles across all screens. - [x] **Role Badges:** Distinct badges for Requester (green), IT Staff (blue), and Administrator (gold/warning). - [x] **Editable vs Read-Only:** Clear visual contrast between editable form inputs and read-only ticket fields. - [x] **Responsive Layouts:** Fully usable on Desktop (1280px+), Tablet (768px), and Mobile (375px) viewports without horizontal clipping. - [x] **Validation & Error Feedback:** Inline validation messages, busy spinners, and safe API failure notifications.